Exploring Fine-Grained Image–Text Alignment for Referring Remote Sensing Image Segmentation

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***Abstract*— Given a language expression, referring remote sensing image segmentation (RRSIS) aims to identify ground objects and assign pixelwise labels within the imagery. One of the key challenges for this task is to capture discriminative multimodal features via image–text alignment. However, the existing RRSIS methods use one vanilla and coarse alignment, where the language expression is directly extracted to be fused with the visual features. In this article, we argue that a "fine-grained image–text alignment" can improve the extraction of multimodal information. To this point, we propose a new RRSIS method to fully exploit the visual and linguistic representations. Specifically, the original referring expression is regarded as context text, which is further decoupled into the ground object and spatial position texts. The proposed fine-grained image–text alignment module (FIAM) would simultaneously leverage the features of the input image and the corresponding texts, obtaining better discriminative multimodal representation. Meanwhile, to handle the various scales of ground objects in remote sensing, we introduce a text-aware multiscale enhancement module (TMEM) to adaptively perform cross-scale fusion and intersections. We evaluate the effectiveness of the proposed method on two public referring remote sensing datasets including RefSegRS and RRSIS-D, and our method obtains superior performance over several state-of-the-art methods. The code will be publicly available at https://github.com/Shaosifan/FIANet.**

***Index Terms*— Fine-grained image–text alignment, referring image segmentation, remote sensing images.**

I. Introduction

REFERRING remote sensing image segmentation (RRSIS) aims to identify the desired ground objects from remote sensing images guided by the corresponding textual description. It can help users to extract specific regions by their particular needs and improve the efficiency for remote sensing analysis [1]. RRSIS plays an important role in many tasks such as land use categorization, typical object identification, urban management, and environmental monitoring [2].

In the past years, deep learning has made great progress in a wide range of remote sensing tasks including super-resolution [3], [4], [5], scene classification [6], [7], [8], visual captioning [9], [10], [11], object detection [12], [13], [14], hyperspectral anomaly detection [15], [16], [17], and semantic segmentation [18], [19], [20], [21]. Unlike traditional remote sensing semantic segmentations, RRSIS simultaneously considers the images and textual descriptions and extracts the specific ground objects under text guidance.

There have been several researches in the field of referring image segmentation for natural images over the past few years. Early works relied on convolutional neural networks and recurrent neural networks to extract visual and linguistic representations that are subsequently fused by simple concatenation to generate pixel-level results [22], [23], [24]. Then, some approaches focus on the elaborate design of image–text alignment to learn more discriminative multimodal representations [25], [26], [27], [28]. More recently, Transformer was introduced in the referring image segmentation task and exhibited superior performance than the prior works [29], [30], [31].

Different from natural images, remote sensing imagery usually covers a wide range of ground objects with diverse spatial scales and orientations. It limits these methods designed for natural images to be directly applied in the RRSIS with satisfactory performance [32]. For this point, in the past year, many researchers have focused on the RRSIS task and established two datasets for remote sensing including RefSegRS [1] and RRSIS-D [32], which promotes the development of the field of RRSIS. In these methods, both cross-scale and intrascale information are utilized to accommodate the unique characteristics of remote sensing images, aligning the images with paired texts to achieve a multimodal representation.

One of the key challenges for this RRSIS task is to learn discriminative multimodal features via text–image alignment. Previous RRSIS methods [1], [32] typically employed one kind of vanilla and coarse alignment (implicit alignment) of image and text features, as shown in Fig. 1(a), where the linguistic representation is directly fused with the visual features

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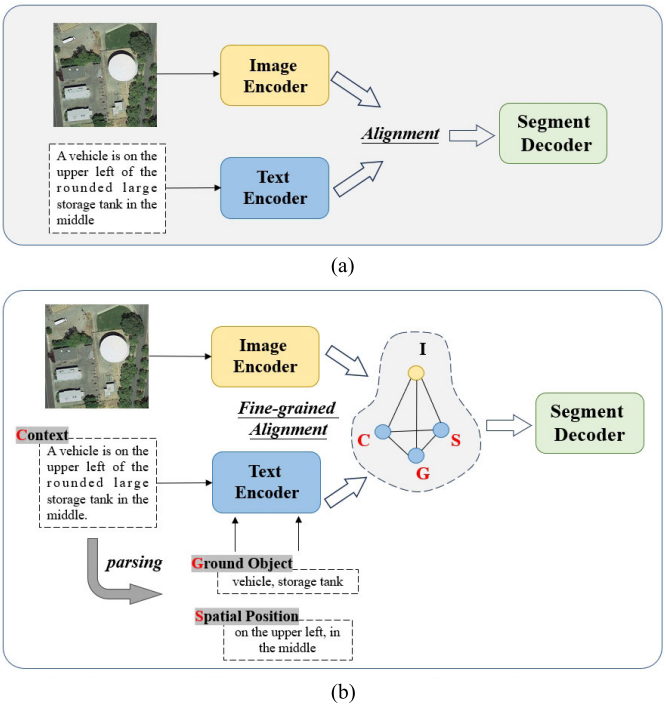


Fig. 1. Motivation of the proposed method. (a) Vanilla image–text alignment employed in the previous referring image segmentation methods for remote sensing. (b) Proposed fine-grained image–text in this article, where the original language expression would be decoupled into ground object fragments and spatial position information. By mining the key elements of images and texts, the association between the image and the referring expression can be clearly constructed, enabling the model to adaptively focus on relevant areas in remote sensing scenarios.

by leveraging pixel-level attention. This is a concise and direct approach, but it neglects the intrinsic information within the referring expression and the fine-grained relationship between the image and the textual description. It might hinder the network from effectively segmentation meeting the complex backgrounds and ground objects with diverse spatial scales in remote sensing.

To handle this issue, we re-examine the vanilla alignment and propose a new paradigm of fine-grained image–text alignment to learn more discriminative multimodal representations. As illustrated in Fig. 1(b), the original referring sentence is regarded as a context expression, and it then is parsed into ground object and spatial position texts. All these sentence fragments will pass a text encoder and obtain fine-grained linguistic representations. By extracting key elements of images and texts, we establish a fine-grained image–text alignment to construct subtle associations between images and their corresponding expressions, enabling the model to adaptively focus on relevant areas in remote sensing scenarios.

In this article, we propose a novel referring image segmentation method for remote sensing, termed FIANet, from the perspective of fine-grained image–text alignment. Specifically, we design a fine-grained image–text alignment module (FIAM) to jointly leverage the features from both input images and the corresponding texts, enabling more discriminative representations across modalities. Meanwhile, to handle the various scales of ground objects in remote sensing, we introduce a text-aware multiscale enhancement module (TMEM), which adaptively performs cross-scale fusion and intersections guided by the texts. We evaluate the effectiveness of the proposed method on two public referring remote sensing datasets, including RefSegRS and RRSIS-D, demonstrating that FIANet achieves superior performance over several state-of-the-art approaches.

The main contributions of this article are summarized as follows.

1) We propose a novel RRSIS method named FIANet. Unlike existing methods, FIANet leverages fine-grained image–text alignment to improve multimodal learning, addressing challenges in handling complex remote sensing scenes. Our method obtains state-of-the-art results on two public remote sensing datasets.
2) We introduce an FIAM to exploit the subtle association between visual and linguistic features, enabling effective segmentations of ground objects under complex backgrounds. Moreover, we design a TMEM to leverage cross-scale multimodal interactions, which can improve FIANet's ability to adapt to ground objects with varying and diverse scales. Comprehensive ablation experiments verify the effectiveness of these designs.

The rest parts of this article are organized as follows. We give a brief description of the background and related work of the RRSIS in Section II. In Section III, we carefully describe our method and the proposed improvements. Many comparative experiments on two public remote sensing datasets and ablation studies are presented in Section IV. Finally, conclusions and future works are drawn in Section V.

II. Background and Related Work

A. Referring Image Segmentation for Natural Images

Referring image segmentation aims to segment a specific target object within an image based on a corresponding textual description, representing a typical multimodal task that has attracted increasing attention. The pioneering work [22] utilizes a convolutional neural network and recurrent LSTM to capture visual and linguistic representation. Liu et al. [23] proposed a recurrent multimodal interaction model that consists of sequential LSTMs to fulfill word-to-image interaction. Margffoy-Tuay et al. [24] designed a modular neural network that divides the problem of referring image segmentation into many subtasks. These methods fuse visual and linguistic representation by simple concatenation to predict pixelwise segmentation output, which constrains the capability of joint learning of images and languages. The subsequent works [25], [26], [27], [28] mainly focus on the elaborate design of image–text alignment to learn more discriminative multimodal representations. Ye et al. [25] introduced a cross-modal self-attention module to learn the long-range relationship between the visual and linguistic features, as well as a gated multilevel fusion module to integrate multilevel self-attentive features. Jing et al. [26] leveraged a cross-model interaction module on the multimodal features by the explicit model of position prior.

Recently, Transformer has exhibited superior performance in the referring image segmentation task [29], [30], [31].

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LAVT [29] employs a vision Transformer [33] as the visual encoder and utilizes an early fusion paradigm to perform hierarchical language-aware visual encoding for capturing multimodal context. Liu et al. [30] designed multimodal mutual attention to better fuse multimodal information, where the features of inputs are extracted by the Swin Transformer and BERT [34], respectively. However, different from natural images, remote sensing imagery usually covers a wide range of ground objects with diverse spatial scales and orientations, which limits the performance of these methods to generate satisfactory segmentation results.

B. Remote Sensing Referring Image Segmentation and Visual Grounding

In the past year, researchers have begun to pay attention to the field of RRSIS, and two datasets, including RefSegRS [1] and RRSIS-D [32], were proposed successively. Yuan et al. [1] tried the first attempt to handle the RRSIS task and proposed a novel language-guided cross-scale enhancement (LGCE) module to improve the results on small ground objects. Liu et al. [32] introduced a rotated multiscale interaction network (RMSIN) to mitigate the issues caused by diverse spatial scales and orientations in the remote sensing imagery, in which intrascale and cross-scale interactions are fully excavated.

Remote sensing visual grounding (RSVG) aims to localize ground objects with bounding boxes referring to the given textual descriptions. Similar to the RRSIS, the one of key challenges for RSVG is to effectively fuse visual and linguistic representations to predict the object's location. Sun et al. [35] established a new visual ground benchmark dataset for remote sensing and proposed a new model composed of image/language encoders and the corresponding fusion module. Furthermore, Zhan et al. [36] introduced a transformer-based method with multilevel cross-modal feature learning to handle large-scale variations and cluttered backgrounds. More recently, Kuckreja et al. [37] proposed a novel grounded large vision-language model that offered multitask capacity for high-resolution remote sensing images. This work can handle multiple tasks simultaneously, including visual grounding, image/region caption, scene classification, and so on.

III. Methodology

A. Overview of the Proposed Method

In this article, we propose a novel RRSIS method named FIANet, which is illustrated in Fig. 2. Similar to the previous works [1], [32], the pipeline of FIANet is divided into four procedures: feature extraction, image–text alignment, multiscale fusion, and segment decoding. Visual and linguistic representations are first extracted from the image and its paired referring expression by an image encoder and a text encoder, respectively. Notably, the original textual description is treated as a contextual expression, which we further decompose into two components: one describing ground objects and the other detailing spatial positions. Thus, three linguistic features are obtained, representing the original contextual expression, ground objects, and spatial positions. Specifically, we employ the natural language toolkit (NLTK) [38] to parse the referring expression based on each dataset's predefined ground object categories. The entire parsing process is conducted offline before training or inference, making it highly efficient. These three linguistic features are extracted by using a pretrained BERT [39].

Algorithm 1 Pseudocode of FIANet in a PyTorch-Like Style

```
# I, T: input image and the corresponding referring text
# FIAM: fine-grained image-text alignment module
# TMEM: text-aware multi-scale enhancement module
# Out: referring segmentation result

# parse the text and extract linguisic features
T_C, T_G, T_S = Sentence_Parsing(T)
F_C, F_G, F_S = BERT(T_C, T_G, T_S)

# visual representation and fine-grain alignment
F_I_0 = I
for i in (1, 2, 3, 4) # the encoder has four blocks
F_I_i = Encoder_Block_i(F_I_i-1)
F_I_i = FIAM(F_I_i, F_C, F_G, F_S)

# multi-scale enhancement with visual/linguisic features
F_I_1, F_I_2, F_I_3 = Downsample(F_I_1, F_I_2, F_I_3)
F_cat = Concat(F_I_1, F_I_2, F_I_3, F_I_4)
F_cat = TMEM(F_cat, F_C)

# obtain final result
Out = Segmenat_Decoder(F_cat)
```

The hierarchical visual features extracted from various stages of the encoder are subsequently aligned with the corresponding linguistic features, thereby enabling the capture of discriminative multimodal representations. For this point, we propose an FIAM to subtly align visual and linguistic representations. After that, the TMEM is implemented to combine these multimodal representations from different levels, which improves the ability of FIANet to adapt to ground objects with varying and diverse scales. Finally, the enhanced multiscale representations would be integrated to generate the pixelwise segmentation by the segment decoder.

Algorithm 1 provides the pseudocode of FIANet in a PyTorch-like style, where the main components of the forward pass are involved. More details about the FIAM and TMEM will be carefully described in Sections III-B and III-C.

B. Fine-Grained Image–Text Alignment

Different from the traditional image–text alignment in the previous methods, we introduce a new multimodal fusion manner from the perspective of fine-grained alignment to capture more discriminative representations. Concretely, given the visual feature $\boldsymbol{F}_I \in \mathbb{R}^{C\times H\times W}$, and the linguistic features $\boldsymbol{F}_C \in \mathbb{R}^{N_C\times D}$ and $\boldsymbol{F}_G \in \mathbb{R}^{N_G\times D}$, and $\boldsymbol{F}_S \in \mathbb{R}^{N_S\times D}$, the FIAM is introduced to perform deep intersections between these visual and linguistic features. Here, C, H, and W denote the number of channels, height, and width of the visual feature maps. Moreover, D is the dimension of word embeddings, and N_C, N_G, and N_S represent the length of context, ground object, and spatial position expressions. The detailed structure of FIAM is shown in Fig. 3. The core components of the FIAM are the object-position alignment

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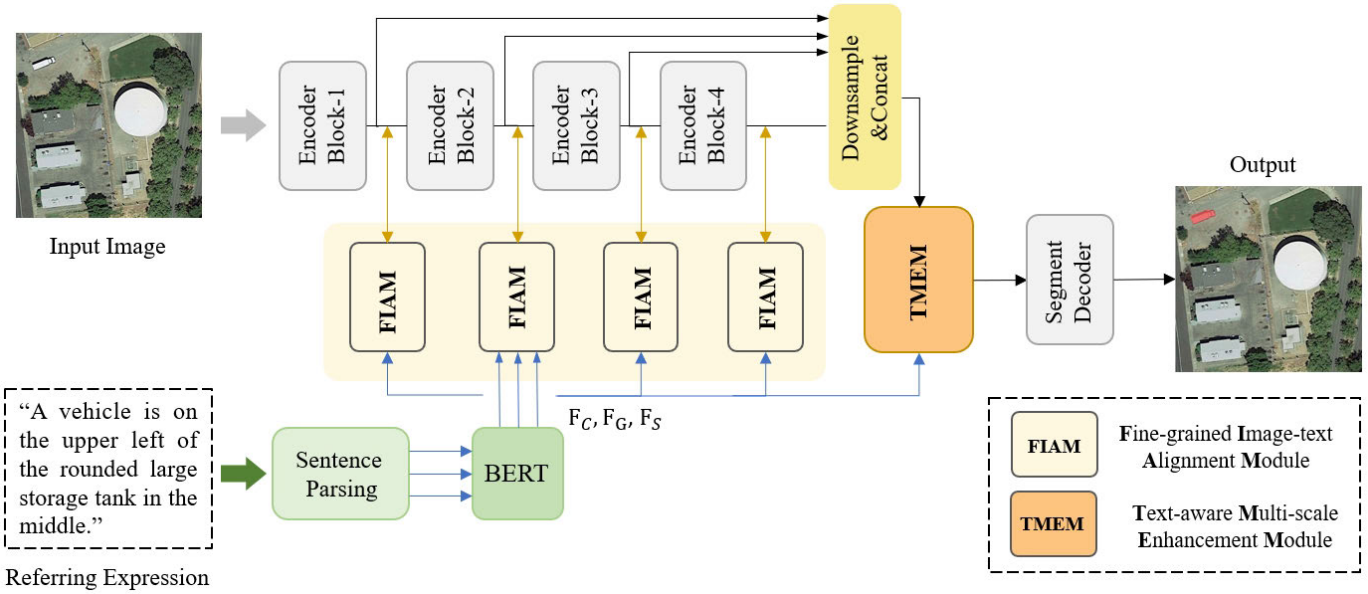


Fig. 2. Framework of the proposed method. The original textual description is regarded as context expression and is further parsed into two fragments about ground objects and spatial positions. There would be three linguistic features in total, including F_C, F_G, and F_S, which denote the representations extracted by the pretrained BERT from the original context expression, ground objects, and spatial positions. FIAMs (see Section III-B) would subtly align visual and linguistic representations, and the TMEM (see Section III-C) is designed to fuse multimodal representations from different levels.

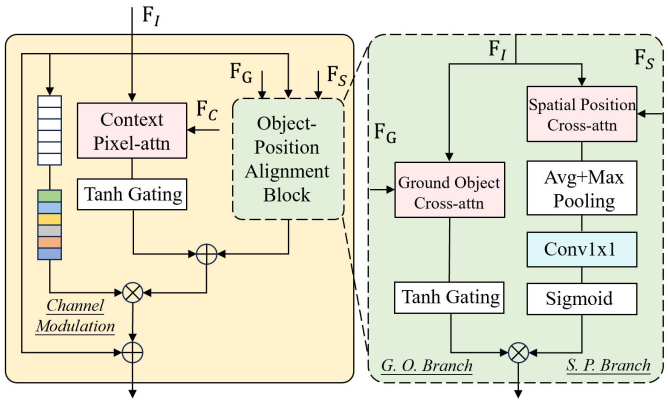


Fig. 3. Illustration of the FIAM that aims to obtain discriminative multimodal representation using visual and fine-grained linguistic features.

block (OPAB), context alignment, and channel modulation, which are carefully described in the following.

1) Object-Position Alignment Block: For each FIAM, we propose an OPAB to perform the deep intersection of features from the ground object and spatial position with the visual representation. This block enables precise alignment of object-related and spatial features and allows the model to capture more accurate relationships between objects and their positions within images, thereby enhancing referring segmentation performance. The detailed structure of OPAB is illustrated in Fig. 3. Specifically, a dual-branch structure is constructed by a ground object branch and a *spatial position branch*. The object branch is established to directly perform multifusion between the textual features of ground objects and the visual features, which can enhance the discriminative ability of the model on the referent target. The main part of the ground object branch is a ground object cross-attention block that can integrate the visual feature $\boldsymbol{F}_I$ and the textual feature $\boldsymbol{F}_G$. Here, we take $\boldsymbol{F}_I$ as the query and $\boldsymbol{F}_G$ as the key and value to achieve feature fusion. This implementation can be defined as

$$\boldsymbol{F}_{IG} = \text{Softmax}\left(\frac{\boldsymbol{F}_I \boldsymbol{W}_q^{IG} \cdot \boldsymbol{F}_G \left(\boldsymbol{W}_k^{IG}\right)^T}{\sqrt{C}}\right) \cdot \boldsymbol{F}_G \boldsymbol{W}_v^{IG} \quad (1)$$

where $\boldsymbol{W}_q^{IG}$, $\boldsymbol{W}_k^{IG}$, and $\boldsymbol{W}_v^{IG}$ are the linear projection matrices that are responding to the query, key, and value, respectively. The C is the dimension of the query.

Moreover, the image-language feature $\boldsymbol{F}_{IG}$ is further modulated by a tanh gate to provide more local details and produce the output $\boldsymbol{F}_{\text{GOB}}$ for this ground object branch. The calculation can be defined as follows:

$$\boldsymbol{F}_{\text{GOB}} = \text{Tanh_Gate}(\boldsymbol{F}_{IG}) \cdot \boldsymbol{F}_{IG} \quad (2)$$

where Tanh_Gate$(\cdot)$ denotes a series of operations, sequentially including the linear projection, the ReLU activation, the linear projection, and the Tanh activation.

The spatial position branch is designed to capture the spatial prior guided by the original visual feature $\boldsymbol{F}_I$ and the textual features of positional description $\boldsymbol{F}_S$. Concretely, $\boldsymbol{F}_I$ and $\boldsymbol{F}_S$ will go through a cross-attention, where $\boldsymbol{F}_I$ is taken as the query and $\boldsymbol{F}_G$ is the key and value

$$\boldsymbol{F}_{IS} = \text{Softmax}\left(\frac{\boldsymbol{F}_I \boldsymbol{W}_q^{IS} \cdot \boldsymbol{F}_S \left(\boldsymbol{W}_k^{IS}\right)^T}{\sqrt{C}}\right) \cdot \boldsymbol{F}_S \boldsymbol{W}_v^{IS}. \quad (3)$$

Then, $\boldsymbol{F}_{IS}$ is input into a series of layers to generate spatial attention, where average and maximum pooling, 1×1 convolution, and sigmoid nonlinearity are implemented. It is shown in Fig. 3 and mathematically described as follows:

$$\begin{aligned} \boldsymbol{F}_{\text{cat}} &= \text{Concat}(\text{Avg_Pool}(\boldsymbol{F}_{IS}), \text{Max_Pool}(\boldsymbol{F}_{\text{IS}})) \\ \boldsymbol{F}_{\text{SPB}} &= \text{Sigmoid}(\text{Conv}(\boldsymbol{F}_{\text{cat}})) \end{aligned} \quad (4)$$

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where $\boldsymbol{F}_{\text{SPB}}$ is the output of this spatial position branch and is regarded as one kind of spatial prior, which involves the multimodal information of visual and textual features. $\boldsymbol{F}_{\text{SPB}}$ is further integrated with ground object features to acquire the final output of OPAB

$$\boldsymbol{F}_{\text{OPAB}} = \boldsymbol{F}_{\text{GOB}} \otimes \boldsymbol{F}_{\text{SPB}} \tag{5}$$

where $\otimes$ denotes the elementwise multiplication. The output of OPAB is the fine-alignment features that simultaneously consider the ground object and the corresponding spatial attention and can help the FIAM obtain more discriminative representation referring to the specific objects.

2) Context Alignment With Visual Features: Apart from the object-position alignment, we also introduce context alignment to capture the global relationships between image and linguistic features. The original referring expression is treated as a contextual description, which contains more contextual information compared to the sentence fragments of the ground object and spatial position. Given the linguistic contextual feature $\boldsymbol{F}_C$ and the visual feature $\boldsymbol{F}_I$, one pixel-attention is employed to combine these two features

$$\boldsymbol{F}_{IC} = \text{Pixel_Attention}(\boldsymbol{F}_I, \boldsymbol{F}_C). \tag{6}$$

Here, the pixel attention is implemented by the pixel-word attention module (PWAM) [29], which aligns the visual representations with the language features of the original description. Similar to the ground object cross-attention and spatial position cross-attention, we use $\boldsymbol{F}_I$ as the query and $\boldsymbol{F}_C$ as the key and value in the pixel attention. Moreover, the image-language feature $\boldsymbol{F}_{IC}$ is also modulated by a tanh gate. The calculation can be defined as follows:

$$\hat{\boldsymbol{F}}_{IC} = \text{Tanh_Gate}(\boldsymbol{F}_{IC}) \cdot \boldsymbol{F}_{IC}. \tag{7}$$

After acquiring $\hat{\boldsymbol{F}}_{IC}$ and $\boldsymbol{F}_{\text{OPAB}}$, the multimodal features $\boldsymbol{F}_{\text{CGS}}$ can further be obtained by combining these two

$$\boldsymbol{F}_{IO} = \hat{\boldsymbol{F}}_{IC} + \boldsymbol{F}_{\text{OPAB}}. \tag{8}$$

3) Channel Modulation: In order to encourage information exchange across channels, we here proposed a channel modulation operator to readjust the extracted multimodal features, which can further enhance the discriminative ability of the proposed method. Specifically, channelwise dependencies can be obtained by

$$\boldsymbol{c} = \sigma(\boldsymbol{W}_2\delta(\boldsymbol{W}_1 \cdot \text{Avg_Pool}(\boldsymbol{F}_{IO}))) \tag{9}$$

where $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are learned weights to perform channel shrink and channel expansion, respectively. δ denotes the ReLU function, and σ indicates the sigmoid function.

Then, the channel weight $\boldsymbol{c}$ would be utilized to recalibrate the multimodal feature F_{IO} to acquire the final output of the FIAM with the original input $\boldsymbol{F}_I$. The calculation is given as follows:

$$\boldsymbol{F}_{\text{FIAM}} = \boldsymbol{c} \otimes \boldsymbol{F}_{IO} + \boldsymbol{F}_I. \tag{10}$$

Overall, through fine-grained image–text alignment, $\boldsymbol{F}_{\text{FIAM}}$ effectively integrates the visual feature with text features at different levels covering the context, ground objects, and spatial positions. Compared to existing methods, the proposed network can acquire more fine-grained informative features, thereby enabling more accurate pixel-level segmentation results.

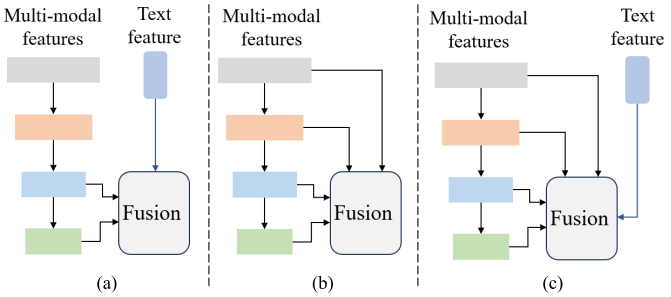


Fig. 4. Comparisons of cross-scale interaction within LGCE [1], RMSIN [32], and our proposed method. Different from these two works, our method can fully explore the multiscale information of visual representations with text features. (a) LGCE. (b) RMSIN. (c) Ours.

C. Text-Aware Multiscale Enhancement

The ground objects in remote sensing images exhibit a wide of scales, hindering the effective extraction of referring objects. Therefore, cross-scale interaction, which leverages features from different scales, plays an important role in this RRSIS task. As shown in Fig. 4(a) and (b), LCGE [1] explores the cross-scale correlation with only two scales assisted by text guidance, and RMSIN [32] employs cross-scale interaction between all the four scales' features without text guidance. These methods have not fully explored the multiscale information of visual representations with linguistic features. Drawing inspiration from these methods, we propose a TMEM to effectively leverage multiscale visual and linguistic features, where the schematic is shown in Fig. 4(c). We compare the proposed TMEM with the cross-scale correlation approaches designed in LCGE and RMSIN, and verify the superior performance of the TMEM over these two approaches. More detailed information can be found in Section IV-C.

As shown in Fig. 5, before inputting into the TMEM, all the multimodal representations from different stages of the image encoder are first preprocessed to ensure that they have the same spatial dimensions. Supposing that F_I^i denotes the output of stage i, these features are downsampled with average pooling to the same size and then concatenated along the channel dimension

$$\begin{aligned} \hat{\boldsymbol{F}}_I^i &= \text{Downsample}\left(\boldsymbol{F}_I^i\right) \\ \hat{\boldsymbol{F}}_{\text{cat}} &= \text{Concat}\left(\hat{\boldsymbol{F}}_I^1, \ldots, \hat{\boldsymbol{F}}_I^4\right) \end{aligned} \tag{11}$$

where $\hat{\boldsymbol{F}}_I^i$ denotes the downsampled features of $\boldsymbol{F}_I^i$ and $\hat{\boldsymbol{F}}_{\text{cat}}$ denotes the concatenated features.

The next question is how to construct a feasible structure of TMEM to achieve the multiscale feature fusion. For this point, we design a concise and effective structure based on transformer decoders to capture long-term dependencies across different scales. Specifically, $\hat{\boldsymbol{F}}_{\text{cat}}$ and linguistic feature $\boldsymbol{F}_C$ are fed into TMEM to perform multiscale fusion. The text-aware multiscale attention in the TMEM is one kind of multiheaded self-attention (MSA) to perform the deep fusion

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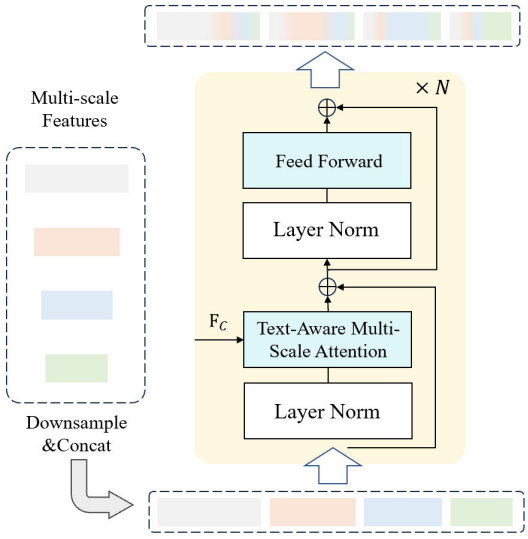


Fig. 5. Illustration of TMEM. Before input into the TMEM, the multiscale features need to be downsampled and concatenated.

and intersection with multimodal features and text guidance. The overall calculation process is given as follows:

$$\begin{aligned} z_0 &= \hat{\boldsymbol{F}}_{\text{cat}} \\ z_i' &= \text{Attention}(\text{LN}(z_{i-1}), \boldsymbol{F}_C) + z_{i-1}, \quad i = 1, \ldots, L_N \\ z_i &= \text{MLP}\big(\text{LN}\big(z_i'\big)\big) + z_i', \quad i = 1, \ldots, L_N \end{aligned} \tag{12}$$

where LN denotes the layer normalization [40] and MLP denotes the multilayer perceptron that has two layers with the GELU nonlinear function [41]. Attention(·) represents the text-aware multiscale attention in the TMEM, and the text representation is integrated into the multiscale fusion to enhance the discriminative ability for referring objects with diverse scales, which can be computed as

$$\hat{z}_{i-1} = \text{Softmax}\left(\frac{\text{LN}(z_{i-1})\boldsymbol{W}_q^{i-1} \cdot \boldsymbol{F}_C\big(\boldsymbol{W}_k^{i-1}\big)^T}{\sqrt{C'}}\right) \cdot \boldsymbol{F}_C \boldsymbol{W}_v^{i-1}. \tag{13}$$

After obtaining the multiscale enhanced features, the output of TMEM is split along the channel dimension and is upsampled to the original spatial dimension. These enhanced multiscaled features are passed through a scale-aware gate [32] and a segment decoder to make the final mask prediction.

D. Implementation Details

In this article, the proposed method is implemented using PyTorch [42]. Following the setting of [1] and [32], we utilize the Swin Transformer as the visual backbone, which is pretrained on ImageNet22K [43], and use the BERT from HuggingFace's Transformer library [44] as the text encoder. The image encoder and the text encoder will be fine-tuned on the remote sensing dataset. Referring to the work [32], we use the combination of cross-entropy loss and dice loss to train our model, where the weight of dice loss is set to 0.1.

There are two RRSIS datasets, including RefSegRS [1] and RRSIS-D [32], and all the images are resized at 480 × 480 pixels. For the RefSegRS and RRSIS-D datasets, we train the model for 60 epochs and 40 epochs, with a learning rate of 5e−5 and 3e−5, respectively. In the training phase, AdamW [45] is adopted to optimize the model, and weight decay is set to 0.1. All the experiments are conducted on an NVIDIA GeForce RTX 4090 GPU with a batch size of 8.

IV. Experiments

A. Dataset and Metrics

In this article, we use two public remote sensing datasets, RefSegRS [1] and RRSIS-D [32], to evaluate the effectiveness of the proposed method. These datasets were recently introduced, contributing to the advancement of the RRSIS task.

1) *RefSegRS [1]:* This dataset contains 4420 image–text–label triplets in total. The training set has 2172 triplets, the validation set has 431 triplets, and the rest 1817 triplets are in the test set. The whole dataset covers 14 categories including road, vehicle, car, van, building, and so on, with five attribute tags used to describe these ground objects. The image size is 512 × 512, and the spatial resolution is 0.13 m.
2) *RRSIS-D [32]:* Compared with the RefSegRS, the RRSIS-D is a larger benchmark and comprises a collection of 17 402 images, masks, and referring expressions, with 12 181 for training, 1740 for validation, and the rest 3481 for testing. RRSIS-D contains 20 categories for the semantic labels and referring expressions, such as airplane, golf field, expressway service area, baseball field, and stadium. The image size in this dataset is 800 × 800 with spatial resolutions ranging from 0.5 to 30 m.

Following some earlier works [1], [29], [32], we employ overall Intersection over Union (oIoU) and mean IoU (mIoU) to evaluate the overall results of different methods. Specifically, oIoU computes the ratio of the total intersection area to the total union area across the entire test set, thereby giving greater weight to large ground objects. The mIoU represents the average IoU computed between the predictions and their corresponding ground truths across all test samples, which treats large and small ground objects equally. Moreover, precisions at threshold values of 0.5–0.9 (denoted as Pr@X) are also utilized to measure the ratio of test images that pass a specific IoU threshold.

B. Comparisons With Other Methods

We compare the proposed method with some state-of-the-art for referring image segmentation on the RefSegRS and RRSIS-D datasets. Among these methods, LGCE [1] and RMISN [32] are specifically designed for remote sensing images, and the others are for natural images. The results of different methods are provided in Tables I–IV. For a fair comparison, we reimplement some state-of-the-art including LAVT [29], CrossVLT [47], LGCE [1], and RMSIN [32], where the total number of train epochs for RefSegRS is set

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TABLE I

RESULTS OF REFERRING IMAGE SEGMENTATION WITH DIFFERENT METHODS ON THE **REFSEGRS DATASET**. THE BEST PERFORMANCE IS SHOWN IN BOLDFACE

Methods	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
LSTM-CNN [22]	ECCV'2016	15.69	10.57	5.17	1.10	0.28	53.83	24.76
ConvLSTM [46]	CVPR'2018	31.21	23.39	15.30	7.59	1.10	66.12	43.34
CMSA [25]	CVPR'2019	28.07	20.25	12.71	5.61	0.83	64.53	41.47
BRINet [27]	CVPR'2020	22.56	15.74	9.85	3.52	0.50	60.16	32.87
LAVT [29]	CVPR'2022	70.23	55.53	30.05	14.42	4.07	76.21	57.30
CrossVLT [47]	TMM'2023	71.16	58.28	34.51	16.35	5.06	77.44	58.84
RMISN [32]	CVPR'2024	71.60	55.97	31.87	11.72	1.93	71.73	57.78
LGCE [1]	TGRS'2024	76.55	67.03	44.85	19.04	5.67	77.62	61.90
FIANet (ours)	—	**84.09**	**77.05**	**61.86**	**33.41**	**7.10**	**78.32**	**68.67**

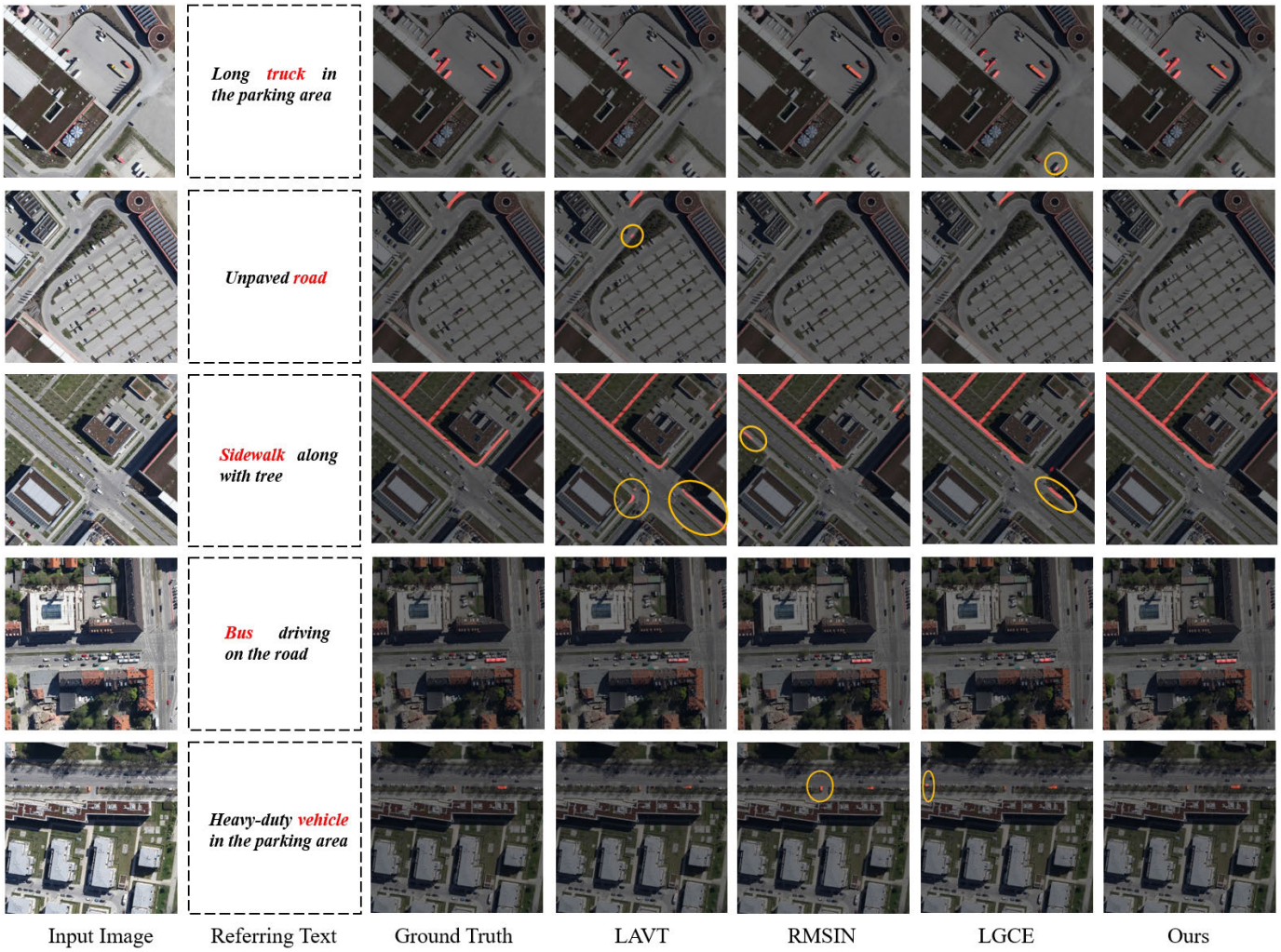


Fig. 6. Qualitative comparisons of different methods on the RefSegRS dataset. The predicted masks are superposed on the original images, and false alarms are circled in yellow. (Best view in Zoom.)

to 60 and the one for RRSIS-D is 40. Meanwhile, for some early published approaches, we take these results reported in LGCE [1] and RMISN [32].

1) Quantitative Results on the RefSegRS Dataset: Table I carefully lists the overall results of different methods on the RefSegRS. It can be seen that our proposed method outperforms other methods across all the metrics on this dataset. Particularly, our method obtains gains of 6.77% in mIoU over the second-best LGCE. To further demonstrate the effectiveness of our method, we provide detailed comparisons of the fine-grained categories. The RefSegRS contains 14 kinds of scenes, and the referring segmentation results of mean IoU for different categories are shown in Table II. The results clearly show that the performance of referring segmentation varies significantly across different ground objects. For instance, "road marking" proves challenging to segment, while

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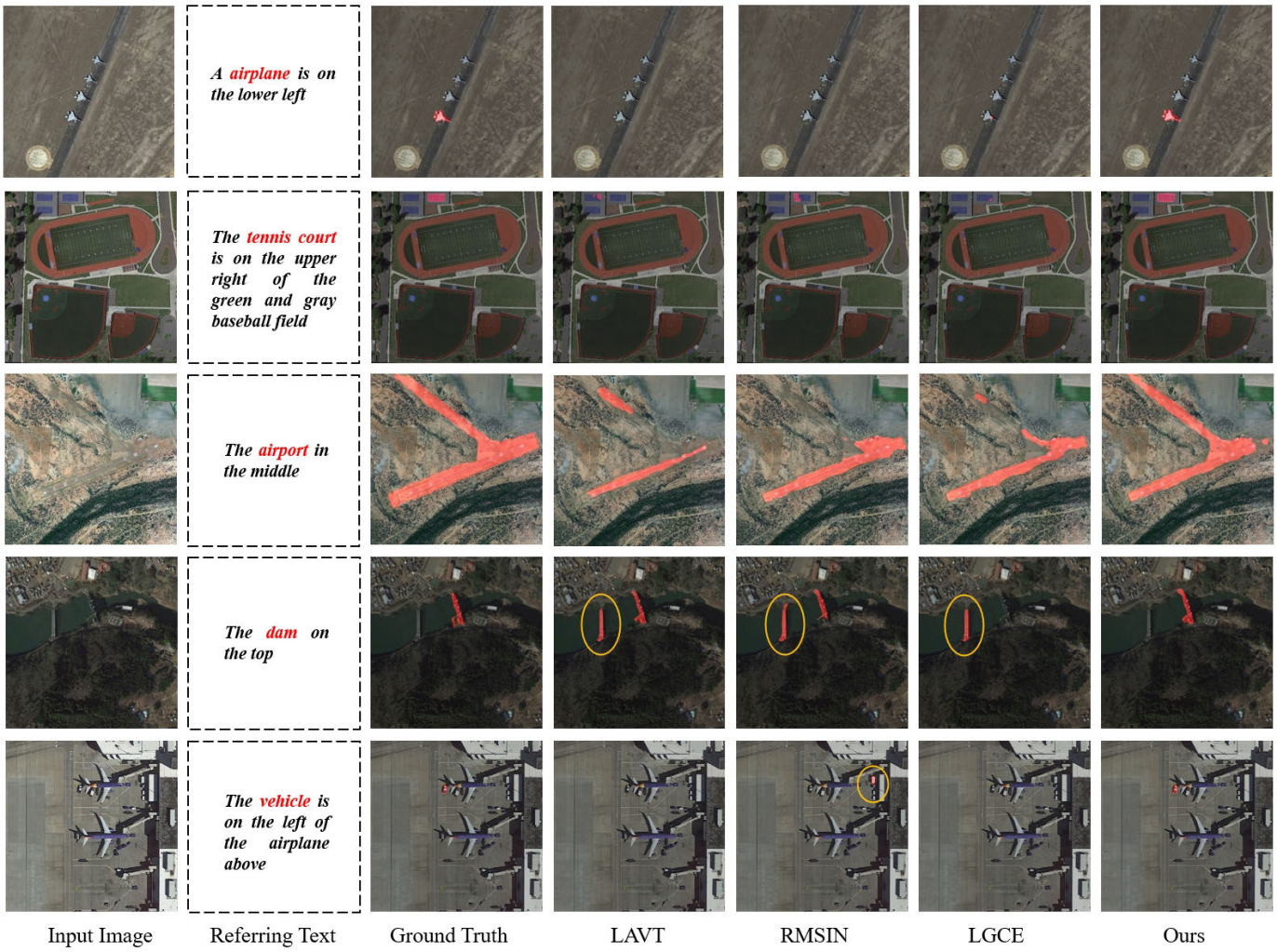


Fig. 7. Qualitative comparisons of different methods on the RRSIS-D dataset. The predicted masks are superposed on the original images, and false alarms are circled in yellow. (Best view in Zoom.)

TABLE II
RESULTS ON EACH CATEGORY OF THE **REFSEGRS DATASET**. THE BEST PERFORMANCE IS SHOWN IN BOLDFACE

category	LAVT	RMSIN	LGCE	Ours
road	70.33	66.67	74.03	**74.13**
vehicle	57.02	58.66	63.05	**70.15**
car	55.13	57.63	60.79	**68.55**
van	38.55	47.09	41.60	**61.06**
building	81.92	76.84	**81.99**	81.34
truck	53.07	51.92	62.69	**74.48**
trailer	44.56	61.65	49.82	**74.92**
bus	52.93	60.20	45.36	**72.40**
road marking	5.74	18.60	6.66	**22.85**
bikeway	50.26	50.35	54.23	**61.16**
sidewalk	57.35	49.12	61.68	**62.90**
tree	57.01	49.82	67.68	**83.75**
low vegetation	41.08	43.73	43.68	**44.84**
impervious surface	81.51	76.55	**83.18**	81.53
average	53.32	54.92	56.89	**66.72**

"impervious surface" is comparatively easier to recognize. In most categories, our method achieves higher mIoU values than LGCE, RMSIN, and LAVT. Furthermore, the average mIoU of our proposed method is substantially higher than that of the other three methods, demonstrating its effectiveness in handling diverse ground objects.

2) Quantitative Results on the RRSIS-D Dataset: Compared to the RefSegRS Dataset, RRSIS-D is a larger dataset with 20 categories of ground objects, providing training samples to optimize the models. The overall results are presented in Table III. Likewise, our method achieves the best performance on this dataset in terms of mIoU, oIoU, and from Pr@0.5 to Pr@0.8. Specifically, the proposed method obtains gains of 1.74% in mIoU over the second-best RMSIN. We have also calculated the segmentation results for each category, as presented in Table IV. Compared to RefSegRS, the ground objects in the RRSIS-D dataset are more challenging to identify due to their diverse and varying scales. Our method obtains the best performance on most ground objects, including road, vehicle, car, van, and so on, and achieves the highest average mIoU with 1.33% higher than the second-best RMSIN.

3) Qualitative Comparisons: We here provide some qualitative comparisons with LAVT, RMSIN, and LGCE on these two datasets. Fig. 6 shows several segmentation results referring to the corresponding texts of the RefSegRS dataset, including

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TABLE III
RESULTS OF REFERRING IMAGE SEGMENTATION WITH DIFFERENT METHODS ON THE **RRSIS-D DATASET**. THE BEST PERFORMANCE IS SHOWN IN BOLDFACE

Methods	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
RRN [46]	CVPR'2018	51.07	42.11	32.77	21.57	6.37	66.43	45.64
CMSA [25]	CVPR'2019	55.32	46.45	37.43	25.39	8.15	69.39	48.54
LSCM [48]	ECCV'2020	56.02	46.25	37.70	25.28	8.27	69.05	49.92
CMPC [49]	CVPR'2020	55.83	47.40	36.94	25.45	9.19	69.22	49.24
BRINet [27]	CVPR'2020	56.90	48.77	39.12	27.03	8.73	69.88	49.65
CMPC+ [50]	TPAMI'2021	57.65	47.51	36.97	24.33	7.78	68.64	50.24
LAVT [29]	CVPR'2022	66.93	60.99	51.71	39.79	23.99	76.58	59.05
CrossVLT [47]	TMM'2023	70.38	63.83	52.86	42.11	**25.02**	76.32	61.00
LGCE [1]	TGRS'2024	69.41	63.06	53.46	41.22	24.27	76.24	61.02
RMISN [32]	CVPR'2024	71.96	65.76	55.16	42.03	**25.02**	76.50	62.27
FIANet (ours)	—	**74.46**	**66.96**	**56.31**	**42.83**	24.13	**76.91**	**64.01**

TABLE IV
RESULTS ON EACH CATEGORY OF THE **RRSIS-D DATASET**. THE BEST PERFORMANCE IS SHOWN IN BOLDFACE

category	LAVT	RMSIN	LGCE	Ours
airport	66.44	68.08	68.11	**68.66**
golf field	56.53	56.11	56.43	**57.07**
expressway service area	76.08	76.68	77.19	**77.35**
baseball field	68.56	66.93	**70.93**	70.44
stadium	81.77	83.09	**84.90**	84.87
ground track field	81.84	81.91	**82.54**	82.00
storage tank	71.33	73.65	73.33	**76.99**
basketball court	70.71	72.26	74.37	**74.86**
chimney	65.54	68.42	**68.44**	68.41
tennis court	74.98	76.68	75.63	**78.48**
overpass	66.17	**70.14**	67.67	70.01
train station	57.02	**62.67**	58.19	61.30
ship	63.47	64.64	63.48	**65.96**
expressway toll station	63.01	**65.71**	61.63	64.82
dam	61.61	68.70	64.54	**71.31**
harbor	60.05	60.40	60.47	**62.03**
bridge	30.48	36.74	34.24	**37.94**
vehicle	42.60	47.63	43.12	**49.66**
windmill	35.32	41.99	40.76	**46.72**
average	62.44	65.13	64.12	**66.46**

TABLE V
ABLATION STUDIES ON THE FIAM AND TMEM

FIAM	TMEM	P@0.5	P@0.7	P@0.9	oIoU	mIoU
		78.37	43.04	2.86	74.90	62.24
✓		83.21	57.29	4.79	77.83	66.68
	✓	80.96	53.99	4.29	75.62	65.39
✓	✓	84.09	61.86	7.10	78.32	68.67

TABLE VI
ABLATION ON EFFECT OF DIFFERENT DESIGNS OF FIAM

C.M.	G.O.B.	S.P.B.	P@0.5	P@0.7	P@0.9	oIoU	mIoU
			80.96	53.99	4.29	75.62	65.39
✓			81.40	53.99	4.73	76.07	65.76
✓	✓		83.27	57.46	5.34	77.40	67.01
✓	✓	✓	84.09	61.86	7.10	78.32	68.67

truck, road, sidewalk, bus, and vehicle scenes, which are marked in red. Moreover, Fig. 7 illustrates the outcomes of the RRSIS-D dataset covering several ground objects such as airplane, tennis court, airport, dam, and vehicle. Some false alarms of different methods are circled in yellow. In addition, it achieves more precise localization of ground objects while reducing false alarms. These visual comparisons highlight the robustness of the proposed method across diverse ground objects and scales, ranging from tiny vehicles to medium-sized dams and large airports.

C. Ablation Studies

We conduct a series of ablation experiments on the test subset of the RefSegRS dataset to validate the effectiveness of the core components of our method.

1) Effectiveness of FIAM and TMEM: We design some experiments to assess the importance of FIAM and TMEM, and the results are listed in Table V. The baseline without FIAM and TMEM leverages a traditional image–text alignment used in LAVT. As shown in Table V, the introduction of FIAM can largely improve the segmentation results, where mIoU obtains an increase of 4.44%. The combination of the FIAM and TMEM further promotes the performance of the proposed method. To demonstrate the effectiveness of these two modules, we visually compare the segmentation maps on some samples of the RefSegRS dataset, as shown in Fig. 8. The results indicate that the proposed method, incorporating FIAM and TMEM, achieves superior performance and improved segmentation outcomes.

2) Effect of Different Designs of FIAM: In order to provide an in-depth understanding of FIAM, we carry out some experiments to explore the effect of different designs of FIAM. For this point, we remove some key components of this module to record the change in metrics. As provided in Table VI, we explore the influences of channel modulation (C.M.), ground object branch (G.O.B), and spatial positional branch (S.P.B). It is obvious that through the integration of these components, the proposed method obtains better performance. Furthermore, this verifies the effectiveness of the fine-grain image–text alignment.

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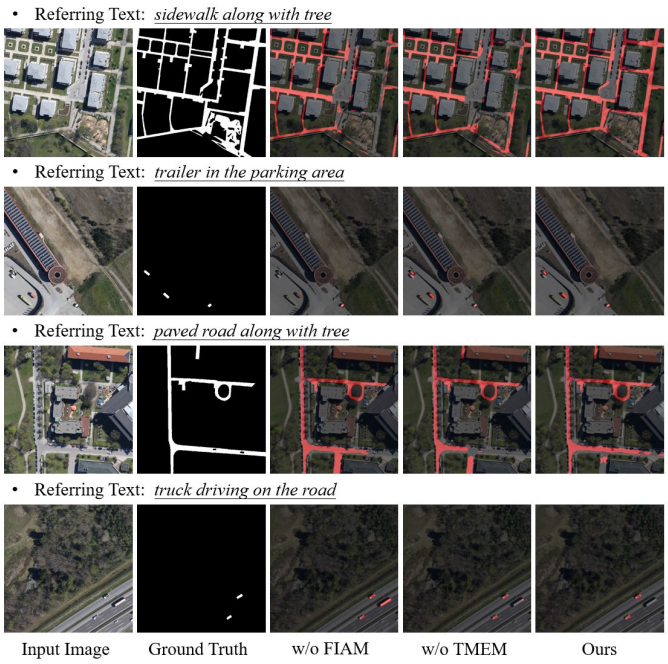


Fig. 8. Qualitative comparisons of different settings on the RefSegRS dataset. (Best view in Zoom.)

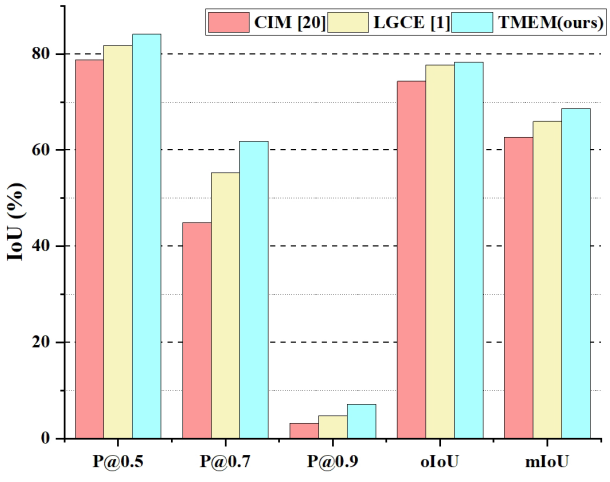


Fig. 9. Comparisons of different designs of multiscale fusion.

3) Effect of Different Designs of Multiscale Fusion: To further demonstrate the efficacy of the proposed TMEM, we here use the other two designs of multiscale fusion to be comparisons, i.e., the cross intersection module (CIM) [32] and LGCE [1]. We use the CIM or LGCE to replace the TMEM, and the other designs remain the same. Fig. 9 shows that the proposed TMEM outperforms CIM and LGCE across all metrics, highlighting the importance of referring text in multiscale feature enhancement and demonstrating the effectiveness of the proposed TMEM.

V. Conclusion

In this article, we propose a new referring image segmentation method for remote sensing, named FIANet, from the perspective of fine-grained image–text alignment. Specifically, we design an FIAM to exploit the subtle association between the visual and linguistic features and learn better discriminative multimodal representations. Moreover, to handle the various scales of ground objects in remote sensing, we introduce a TMEM to adaptively perform cross-scale fusion and intersections under text guidance. We evaluate the effectiveness of the proposed methods on two public referring remote sensing datasets, including RefSegRS and RRSIS-D, demonstrating that our method achieves superior performance over several state-of-the-art methods. Meanwhile, comprehensive ablation experiments also verify the effectiveness of FIAM and TMEM.

While the proposed method achieves promising results in RRSIS, there remains significant room for further exploration in this task. Future work could focus on developing more efficient multimodal fusion strategies between image and linguistic features to enhance applicability in practical scenarios. In addition, foundation models have demonstrated great potential in computer vision and remote sensing tasks, making their integration into this task a valuable direction for future research.

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